WASP-3b

R-1095

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-3_150504` · created 2026-07-30T18:23:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457146.8078 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.144 +/- 0.042
Transit depth (Rp/Rs)^2	2.06 +/- 1.21 [%]
Orbital Inclination (inc)	80.8 +/- 2.86
Airmass coefficient 1 (a1)	0.534 +/- 0.09
Airmass coefficient 2 (a2)	0.247 +/- 0.055
Scatter in the residuals of the lightcurve fit is	17.92 %
Best Comparison Star	#3 - [292, 207]
Optimal Aperture	25.44
Optimal Annulus	48.72
Transit Duration (day)	0.052 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.144 ± 0.042 vs 0.10538 (deviation 0.78σ)
Duration fit vs published	0.052 d vs 0.1167 d (deviation 2.74σ)
Mid-time O–C	3.9 min
Depth SNR	2.9
Residual scatter	17.92 %

Light curve

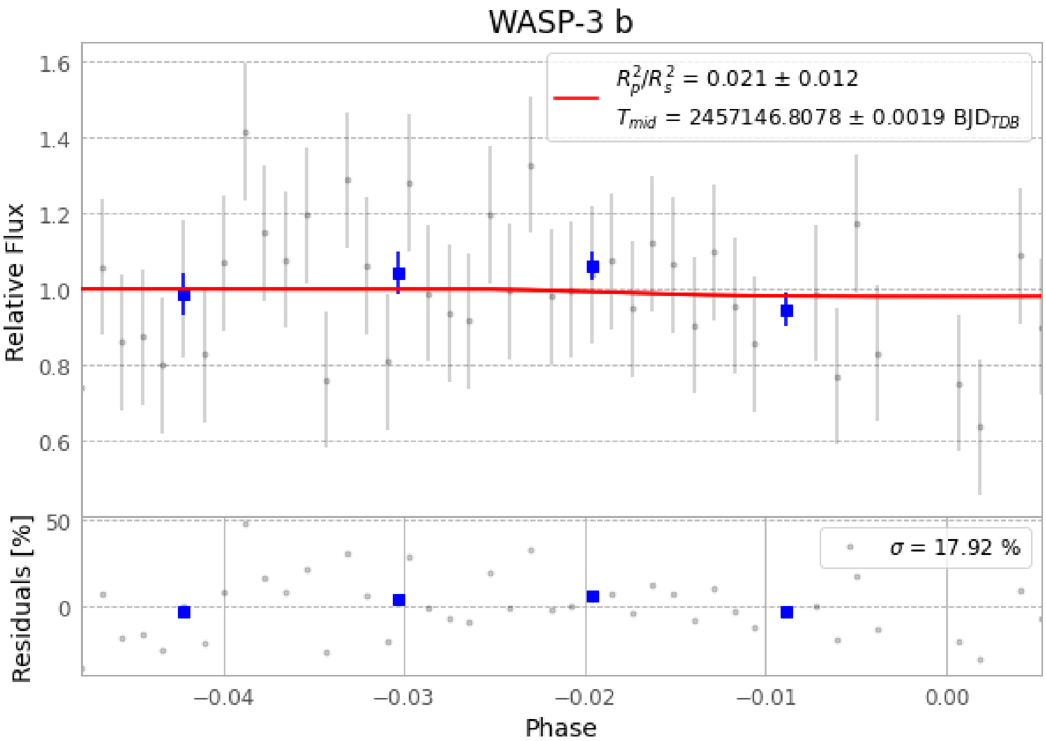

Figure 1 — FinalLightCurve_WASP-3 b_2015-05-03.png (SHA-256 a2c74d48d716fe83…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [548, 232], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 8d40ec43ff1c6d4e…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

86 FITS frames (56 MB), `WASP-3150504051212.FITS` → `WASP-3150504092712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-150504.log (407e7c95fc4b…), FinalLightCurve_WASP-3 b_2015-05-03.png (a2c74d48d716…), FinalLightCurve_WASP-3 b_2015-05-03.csv (388871811c39…)

Compute resources

Wall time 395.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)